# 1. PROBLEM STATEMENT

Workers in high-risk industries such as construction, oil and gas, mining, and manufacturing are regularly exposed to hazards including falling objects, collisions, and toxic gases. Despite advancements in safety regulations and protective equipment, life-altering injuries continue to occur due to limited hazard awareness and delayed emergency response. These challenges show a need for a more intelligent monitoring and alert system to better protect workers on the job.

## 1.1. Need Statement

In 2023, there were 5,283 fatal work injuries recorded in the United States in 2023 [1]. Construction had the most fatalities among all industry sectors in 2023 with a total of 1,075. This was the highest fatality rate for the sector since 2011 [1]. According to the Bureau of Labor Statistics, hard hats were worn by only 16% of workers who sustained head injuries on the job [2]. Toxic gas presents another hazard. According to APnews, 14 workers were hospitalized due to carbon monoxide poisoning even though they had ventilation on the site of construction [3]. Another health issue is presented by UV radiation. In Australia, for example, a large number of construction workers were exposed daily to solar UVR; solar UV was identified as the most common carcinogen in that industry [4]. These statistics reveal the persistent risks faced by workers in hazardous industries, highlighting how current safety gear and practices fall short in preventing serious injuries and fatalities. Cases such as these show the need for smart sensing PPE to better protect workers.

## 1.2. Objective Statement

SmartHat is a fully integrated smart hard hat designed to enhance worker safety through built-in sensing and alert capabilities. It continuously monitors carbon monoxide levels, impacts, high UV exposure, and abnormal vital signs, while providing real-time alerts to the user and those nearby. By enabling early detection and rapid response to potential dangers, the SmartHat helps reduce the risk of accidents while supporting greater operational efficiency and protection on the job site.

## 1.3. Background and Related Work

The concept behind the SmartHat is to transform a traditional hard hat by integrating advanced sensors, creating a comprehensive safety solution for workers in hazardous environments. Although there are devices with some functionality, no product on the market today provides the entire range of capabilities intended for SmartHat. Another product, Guardio, has created a smart helmet that has fall and impact detection capabilities [5]. Its capabilities are limited, though, in contrast to SmartHat's wider range of safety and monitoring capabilities. Also, SmartHat seeks to provide an interface where managers and safety representatives can monitor workers' vitals and working conditions. In the event of an emergency, this will enable them to quickly identify health risks and coordinate an immediate response. Although there are already patents on the concept of a smart hard hat, SmartHat aims to offer real-time data via an interactive system and a wider range of safety.

# 2. DESIGN REQUIREMENT SPECIFICATIONS

The following section outlines the requirements, constraints, and standards that are taken into consideration for the design, testing, and implementation of SmartHat.

## 2.1. Requirements

SmartHat will satisfy the requirements of both workers in dangerous environments and safety representatives who oversee their well-being. The requirements are separated into two groups: marketing requirements, which articulate needs that are directly related to customers, and engineering requirements, which highlight the quantifiable performance standards derived from those needs. When combined, they provide a clear benchmark for usability, performance, and safety.

### 2.1.1. Marketing Requirements

The marketing requirements for SmartHat are outlined as such:

1. SmartHat sensors reliably detect impacts and monitor carbon monoxide and vitals, providing real-time alerts.
2. SmartHat includes an auto-adjusting headlamp and UV detection.
3. SmartHat runs on a rechargeable, long-lasting battery designed for all-day use.
4. SmartHat is designed for comfort and safe, everyday wear.
5. SmartHat is impact-resistant and water-resistant, built to withstand tough environments.

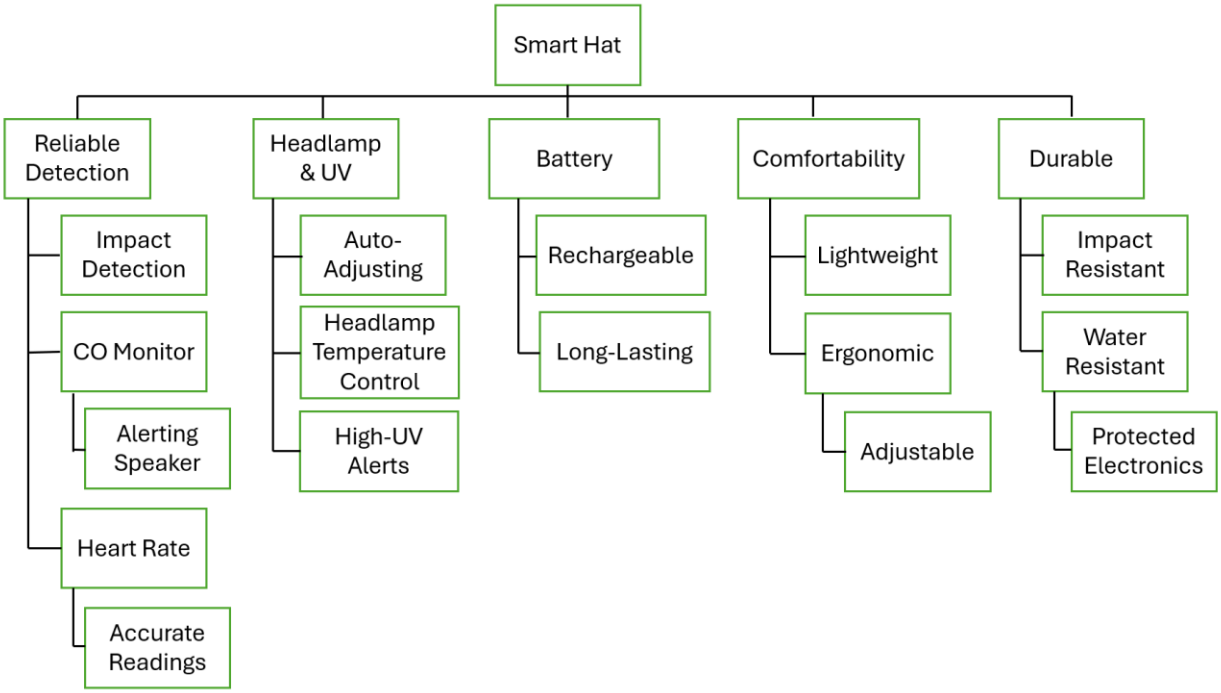


**Figure 2-1. Objective Tree for Smart Hat**

The objective tree expands SmartHat's marketing requirements into easy-to-read categories. The engineering design requirements further discuss the technical design plan to achieve the marketing requirements.

### 2.1.2. Engineering Requirements

SmartHat must meet clear, measurable engineering requirements to ensure it meets workers' and safety staff's needs in hazardous industries. Table 2-1 summarizes these requirements.

**Table 2-1: Engineering Design Requirements**

| Marketing Requirements | Engineering Requirements | Description |
|---|---|---|
| 1 | The CO sensor must detect concentrations ≥ 50 ppm within 5 seconds, with accuracy ±5 ppm. | Ensures timely detection of dangerous carbon monoxide levels and reliable alert triggering. |
| 1 | The impact detection system detects impacts ≥ 35 g. | Concussive forces usually weigh more than 50g. |
| 1 | The vitals sensor must monitor the heart rate with an accuracy of ±3bpm and send an alert if the user heart rate surpasses 180bpm. | Ensures reliable monitoring of worker vitals for early health risk detection. |
| 2 | The headlamp must maintain a brightness of ≥ 200 lumens and automatically activate when ambient light is < 50 lux. | Guarantees adequate visibility in dark environments without manual switching. |
| 3 | The system must operate continuously for ≥ 8 hours on a single charge. | Provides sufficient power for a full work shift without interruption. |
| 4 | The total mass of SmartHat, including electronics, does not exceed 3.2 lbs. | Commercial hard hats average 0.75-1.2 lbs.; 3.2 lbs. maintain comfort and ergonomic safety during prolonged wear. |
| 5 | The enclosure achieves at least IP-54 ingress protection. | IP-54 ensures resistance to dust and splashing water, suitable for outdoor and industrial environments. |

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SmartHat integrates multiple safety features to address the leading risks in hazardous work environments. Its carbon monoxide (CO) sensor detects concentrations of ≥ 50 ppm within 5 seconds, a threshold aligned with OSHA's Permissible Exposure Limit of 50 ppm for an 8-hour TWA [6].

To mitigate risks of head injury, SmartHat employs an impact detection system capable of identifying accelerations ≥ 35gs. According to research on wearable impact sensors, thresholds between 4 and 6 g are frequently used to identify mild shock events, while thresholds between 10 and 20 g are used for verification of severe impacts [7], [8]. By prioritizing serious events like falls or collisions, a detection threshold of 35 g lowers false positives while preserving responsiveness to potentially concussive impacts.

The integrated auto-adjusting headlamp outputs ≥ 200 lumens and activates automatically below 50 lux, providing reliable illumination in dim conditions. Industry standards and commercial helmet-mounted headlamps commonly fall within the 150–300 lumen range, confirming that 200 lumens are sufficient for construction and industrial tasks [9], [10].

SmartHat will also include a heart-rate monitoring system with a sensor monitoring the heart rate within an accuracy of ±3bpm, which will send an alert if the users heart rate surpasses 180bpm. This ensures continuous monitoring of worker vitals, enabling early detection of health risks such as overexertion or heat stress.

The system's battery life will be able to run for at least 8 hours on a single charge, sufficient for a full work shift.

## 2.2. Constraints

The design constraints for SmartHat are described in Table 2-2 below. These constraints pertain to reliability, health and safety, comfortability, and economic factors.

**Table 2-2: Constraints**

| Type | Name | Description |
|---|---|---|
| Economic | Cost | The total budget of the project is $1000. |
| Economic | Time | SmartHat must have 5 completed subsystems by April 2026, restricting the development lifetime to a maximum of 9 months. |
| Energy | Battery | SmartHat must have a minimum battery life of at least 8 hours to support continuous wear during a workday. |
| Health and Safety | Durability | The product must be able to withstand concussive impacts with all sensors and microcontrollers intact and functional. |
| Ergonomic | Duration | SmartHat will be worn by the customer during their entire workday and, therefore, must be lightweight and comfortable to prevent strain and injury. |

An allotment of $1000 dollars has been provided by the Department of Electrical and Computer Engineering for the development of SmartHat and must not be exceeded.

SmartHat must exhibit 3 working subsystems out of the total 5 by the end of the first semester and must be a fully functioning prototype by the end of the second semester. Time must be managed effectively in order to meet these deadlines, and all members of the development team must be aware of and in agreement with the project schedule.

SmartHat is to be worn by the customer for the duration of their workday. Therefore, its battery life must be able to support at least 8 hours of continuous use.

SmartHat is designed to detect impacts on the wearer’s head and must remain structurally intact under such conditions. Failure to maintain integrity would compromise its primary function. Therefore, the durability of SmartHat is of upmost importance.

SmartHat is intended to be worn continuously by any person in a hazardous work environment. Therefore, SmartHat must be lightweight enough to be comfortably worn for many hours without causing strain on the customer.

## 2.3. Standards

Table 2-3 provides three specific standards SmartHat will abide by, providing user and equipment safety.

**Table 2-3: Engineering Standards**

| Specific Standard | Standard Document | Specification / Application |
|---|---|---|
| ANSI Z89.1 | The product meets American National Standard for Industrial Head Protection [ 11]. | SmartHat meets testing requirements including impact resistance, penetration resistance, and provides electrical protection. |
| NEMA 4 | This product follows the International Electrotechnical Commission Ingress Protection Standard [12]. | SmartHat follows the NEMA 4 enclosure standard to protect from dust and water, ensuring a working product through harsh environments. |
| 3GPP Release 13 | This product follows 3rd Generation Partnership Project Cellular standards [13]. | SmartHat will follow the 3GPP Release 13 protocol for cellular communication providing real-time SMS alerts. This standard ensures SmartHat will provide SMS alerts while being in remote areas without access to Wi-Fi. |

SmartHat will use these standards to provide safety to the user, protect internal components, and follow the standard for SMS over cellular networks. This ensures SmartHat will be reliable in real-world environments by meeting standards for safety, durability, and cellular communication.

## 2.4 REFERENCES

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